AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM112
Introduction to Computing and Information Technology

Final Examination
Spring Semester 2025/2026

Date: 10 June 2026

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 3 hours

Instructions:

- Total Marks: **100**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.

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Part I: Multiple Choice Questions (20 marks).

1. What does IoT stand for?
   a. Internet of Technology
   b. Internet of Things
   c. Internal Operating Tool
   d. Interconnection of Terminals

2. What is the primary purpose of the **TRIM command** in SSDs?
   a. To defragment the drive for better performance
   b. To inform the SSD which data blocks are no longer needed
   c. To permanently delete all data using a magnetic field
   d. To create a bit-for-bit copy of the drive

3. In wireless communication, the **inverse square law** states that signal power is proportional to:
   a. $1/d$
   b. $1/d^2$
   c. d^2
   d. d

4. Which of the following is a **symmetric encryption algorithm**?
   a. RSA
   b. Diffie-Hellman
   c. AES
   d. SHA-256

5. A **botnet** is best described as:
   a. A single computer infected with ransomware
   b. A network of compromised computers controlled remotely
   c. A type of antivirus software
   d. A hardware security module for encryption

6. In digital forensics, **slack space** refers to:
   a. The space between tracks on a hard disk
   b. Unused space in the last cluster of a file
   c. Memory allocated for virtual memory
   d. The area where the operating system is stored

7. Which cloud service model provides **virtual machines and storage** to customers?
   a. Software as a Service (SaaS)
   b. Platform as a Service (PaaS)
   c. Infrastructure as a Service (IaaS)
   d. Network as a Service (NaaS)

8. A **collision** in hashing occurs when:
   a. Two different inputs produce the same hash
   b. The same input produces two different hashes
   c. The hash is longer than the original message
   d. The hash cannot be computed due to memory limitations

9. Which of the following is an **advantage of SSDs over HDDs**?
   a. Lower cost per gigabyte
   b. Higher storage capacity
   c. No moving parts, making them more resistant to shock
   d. Easier data recovery after deletion

10. In the context of authentication, **two-factor authentication** requires:
    a. Two different passwords
    b. Something you know and something you have
    c. Two different usernames
    d. Biometric verification only

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Part 2: Short questions (30 marks)

Question 1: (6 marks)
Explain the Advantages of using short-range communication

Question 2: (6 marks)
a. Calculate the **denary (decimal) value** of the binary number **1101.101**. (3 marks)
b. What fraction of the power would be received by the antenna at d = **5**? (3 marks)

Question 3: (6 marks)
List and briefly explain **three types of malware** discussed in the course.

Question 4: [6 marks]
What is the **CIA Triad** in information security? Define each component and give a real-world example of a violation for each.

Question 5: [6 marks]
Explain what does **digital certificates** usually include and the role of a **Certification Authority (CA)** . Why are they necessary for secure web communications?

Part 3: Problem Solving / Long Questions (50 marks)

Question 1: (10 marks)
Mention and describe the three crucial properties of Hashing?

Question 2: (10 marks)
How to Permanently Delete Data from an HDD?

Question 3: [10 marks]
What are the advantages of having Multiprocessors?

Question 4: [10 marks]
a) Explain the difference between **authentication** and **authorization** in computer security. (4 marks)
b) Explain what can botnets be used for. (6 marks)

Question 5: [10 marks]
Describe two methods used in error detection and correction and give an example on each of them

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End of Questions

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